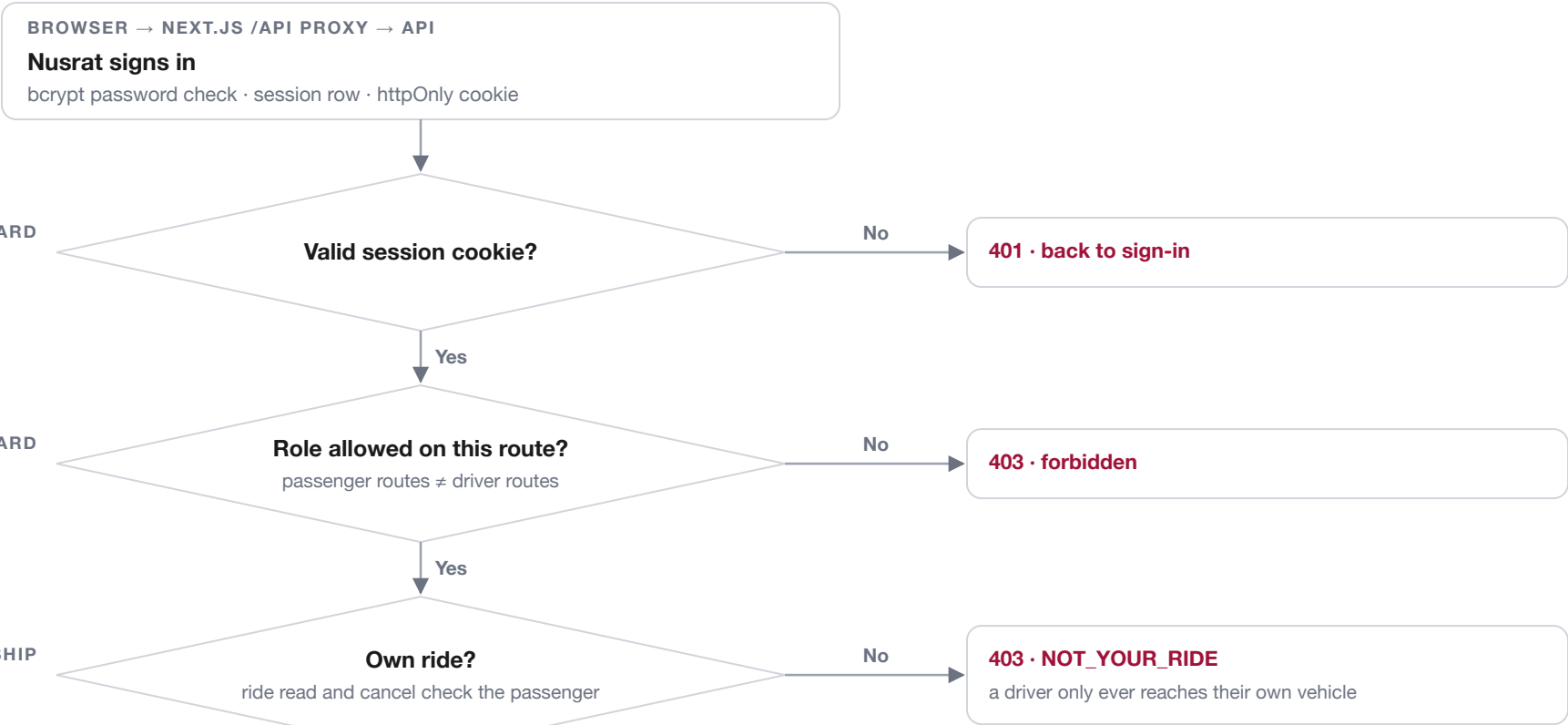


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · read top to bottom · side notes cover concurrent and edge cases

1 SIGN IN



Sign up

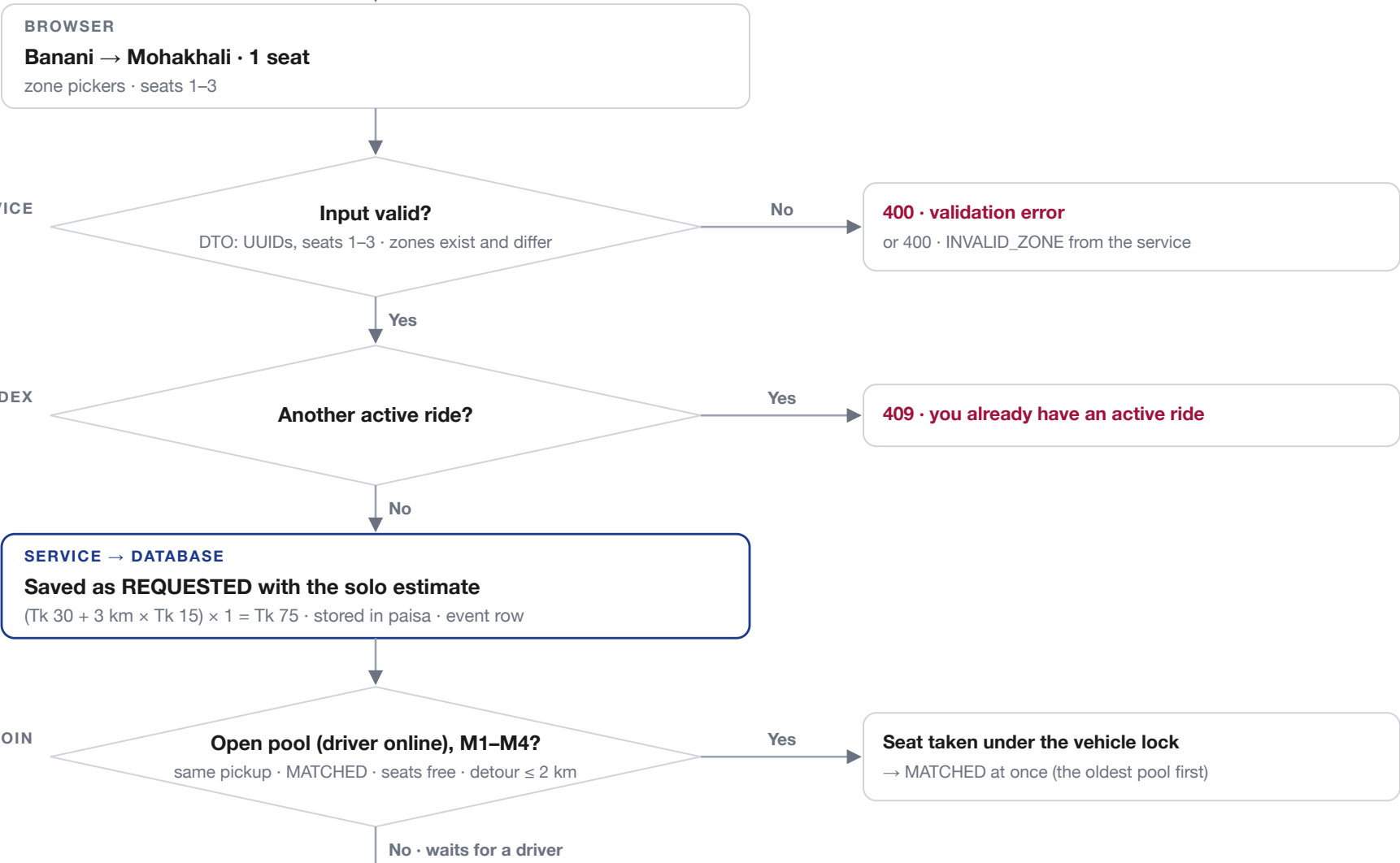
Passengers only; drivers are seeded

Unique email (stored lower case)

Password hashed with bcrypt

Sign-up and login: 5 per minute

2 REQUEST A RIDE



Fare model

(Tk 30 + km × Tk 15) × seats

–20% if 2+ passengers at start

Estimate = solo fare, never exceeded

Discount counts passengers, not seats

Detour rule (M4)

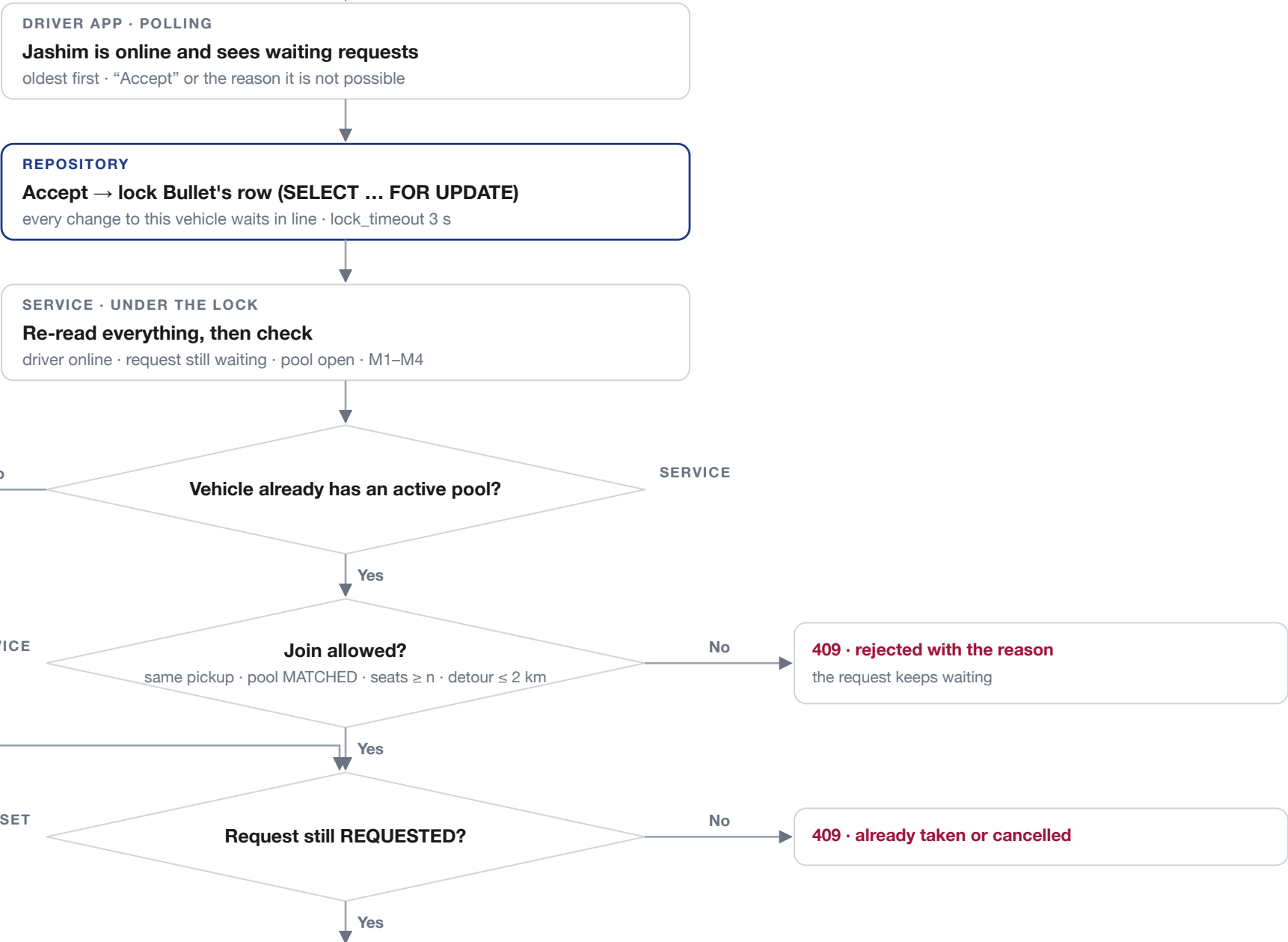
Drop-offs nearest first

detour = in-car km – direct km

Rafiq via Mohakhali: 6 – 4 = 2 km ✓

Banani → Uttara: 16 – 12 = 4 km ✗

3 MATCHING



Last seat, many riders at once

Nusrat holds 2 of Bullet's 3 seats

8 riders ask for the last one

All wait for Bullet's lock

Exactly 1 wins · 7 keep waiting

No pool yet, two accepts at once

First accept creates the pool

Second waits, then joins it

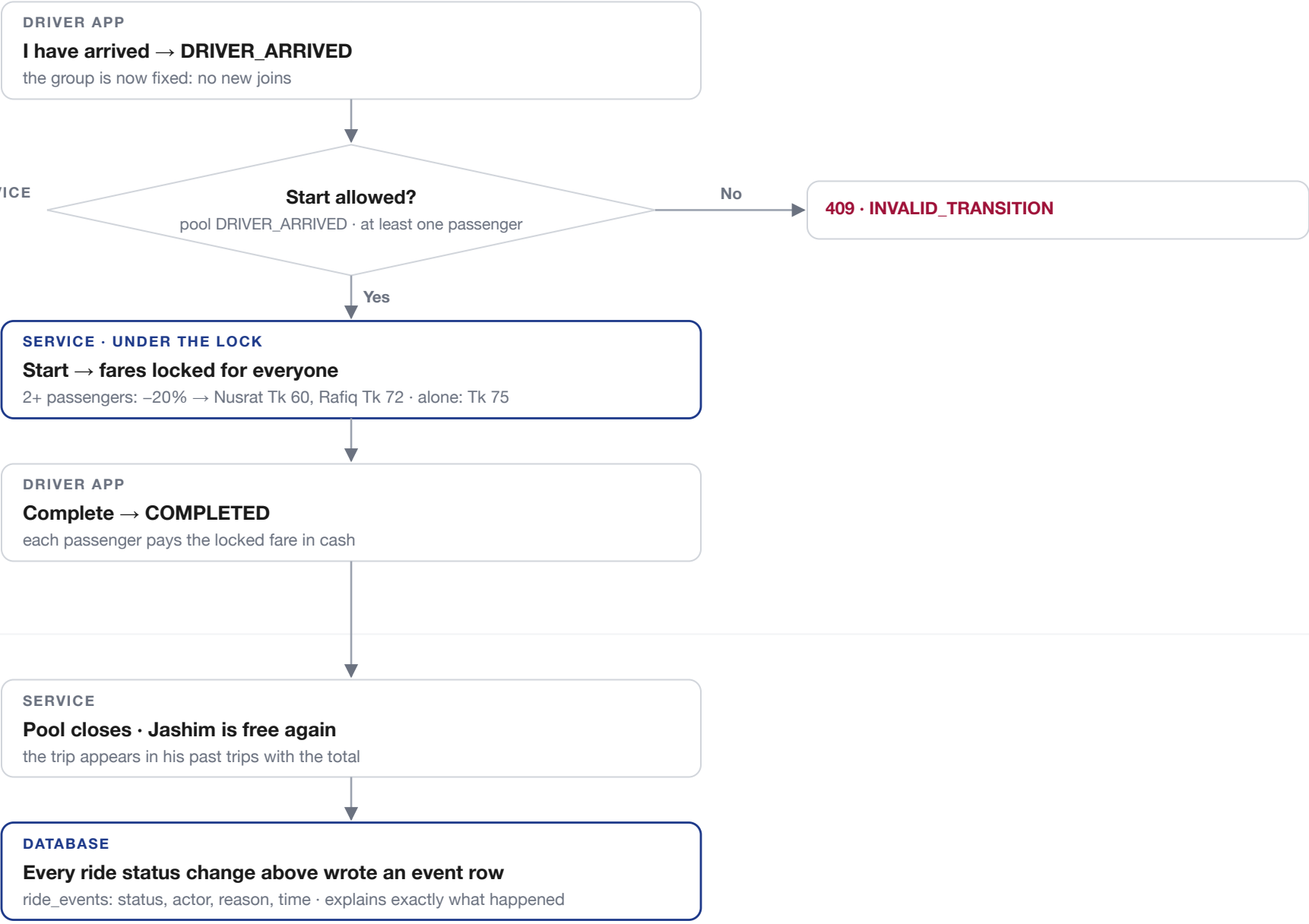
One pool, two passengers

Retries and timeouts

Lock wait > 3 s → 503 BUSY

Auto-join is best effort: if busy, the ride just keeps waiting

4 THE TRIP



Database guarantees

seats_taken ≤ seat_capacity

1 active pool per vehicle

1 active membership per ride

1 active ride per passenger

Online / offline

Changed under the vehicle lock

Offline refused while a pool is active

Accept refused while offline

What the passenger sees

Polls her ride every 3 s (history 10 s)

Own status and own fare only

Driver + vehicle name

Co-riders' first names only

Cancellation rules

Passenger: before the start

→ seats freed at once

Last one out → pool closes

Driver: before the start → passengers back to REQUESTED

After the start: no cancel (409)

5 END OF RIDE